BIOSYS-D-26-00689 (BIO 105919) — Pre-publication QA/QC Ledger

Manuscript: *Robust error-minimization in the genetic code across physicochemical metrics and variant codes: a graph-theoretic analysis in GF(2)⁶* **Authors:** Paul Clayworth, Sergey Kornilov **Status:** In press, *BioSystems* (Elsevier). Article reference BIO_BIOSYS-D-26-00689; production reference BIO 105919. **Prepared:** 15 August 2026 **Corresponding author:** Sergey Kornilov sergey@biostochastics.com **Repository:** https://github.com/biostochastics/codontopo **Release tag:** `v0.6.1` — the single consolidated release accompanying this ledger.

Purpose of this ledger

Between acceptance and typesetting the authors carried out a pre-publication QA/QC audit of every figure, table, and inferential quantity across the manuscript and the supplement. This ledger is the diff manifest for that audit. It records every user-visible change, sorted by manuscript location, with the pre-fix value, the post-fix value, and the justification.

All headline scientific conclusions of the accepted manuscript are unchanged. Four SUPPORTED claims — cross-metric coloring optimality, per-table preservation, ρ-robustness, and topology-avoidance depletion — the EXPLORATORY tRNA-enrichment classification, the FALSIFIED KRAS–Fano prediction, and the three REJECTED / two TAUTOLOGICAL entries are as reported in the accepted version. The claim hierarchy count remains 15 (4 SUPPORTED / 5 EXPLORATORY / 1 FALSIFIED / 3 REJECTED / 2 TAUTOLOGICAL).

Two substantive computational corrections were made during the audit and their effects on individual displayed values are documented explicitly below (§MIS and §NULL). Several displayed p-values shifted at the third and fourth decimal as a result; none crossed a decision threshold. In addition, a number of clarifications, framing corrections, and editorial fixes were applied across the manuscript, supplement, response letter, CHANGELOG, and public README, and the delivery inventory was regenerated from a single source-of-truth.

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§MIS — Maximal-Independent-Set enumeration for tRNA enrichment

Root cause. The initial Bron–Kerbosch pivoting was applied to the conflict graph G rather than to its complement $\bar{G}$. An independent set in G is a clique in $\bar{G}$, so the pivot must be chosen in $\bar{G}$. Fixing the pivot produced the complete MIS enumeration below.

ID	Change	Old value	New value	Justification
MIS-01	Bron–Kerbosch pivot moved to complement graph	Enumerates 2 MIS of size 6 (worst-case Stouffer $p = 0.045$)	Enumerates 332 MIS of size 6; median $p = 0.037$, best 0.012, worst 0.104; 264/332 (79.5%) sets below 0.05	An independent set in G is a clique in $\bar{G}$; pivot must be selected in $\bar{G}$, not in G. Present numbers reflect the post-audit tRNA vector reconciliation (§MIS-11) applied to the complete MIS enumeration.
MIS-02	Topology-breaking-subset test surfaced	Present in `output/trna_evidence.json` but not cited in main text	Explicitly reported in §3.6 and §S10.3: $n = 4$ pairings (yeast mito Thr, *Scenedesmus* mito Leu, *Pachysolen* Ala, *Candida* Ser), Stouffer $p = 0.387$ (null)	The subset is the direct mechanistic test of the compensation-for-disconnection hypothesis; suppressing it would misrepresent the mechanistic evidence.
MIS-03	tRNA-enrichment claim reclassified	Status: SUGGESTIVE, evidence $p = 0.045$ (worst-case MIS)	Status: EXPLORATORY, evidence median MIS $p = 0.037$, worst 0.104; topology-breaking subset $p = 0.387$ (null)	The strict worst-case criterion is not met, and the mechanism-linked subset is null; the signal is retained

ID	Change	Old value	New value	Justification
				as heterogeneous accommodation-of-decoding rather than compensation-for-disconnection.
MIS-04	Main-text Table 7 templated from stats	All rows literal or partly templated	All five rows (Fisher+Stouffer, median MIS, worst MIS, best MIS, topology-breaking subset) render from `stats.trna.*` fields	Prevents future drift between prose and table; the same fields populate the corresponding supplement rows and Table S1 claim justification.
MIS-05	Main-text §3.6 tRNA paragraph rewritten	Two-point summary and stale numbers	Distribution across 332 MIS of size 6 with median/best/worst and 264/332 fraction; topology-breaking-subset null called out	Presents the distribution rather than a two-point summary; discloses the null subset explicitly.
MIS-06	Supplement §S10.2 (MIS) rewritten	Two MIS reported	332 MIS reported; best/worst pairings identified; topology-breaking-subset subsection added	Same as MIS-05 for the supplement.
MIS-07	tRNA legacy repertoires reconciled against primary literature (per pre-publication audit)	*S. cerevisiae* mitochondrial and *Yarrowia lipolytica* mitochondrial counts inconsistent with primary sources	*S. cerevisiae* mitochondrial: 24 tRNAs (2 Arg tRNAs decoding AGR + CGN) after Bonitz *et al.* 1980 (PNAS 77:3167). *Y. lipolytica* mitochondrial: 27 tRNAs (no CGN-Arg) after Kerscher *et al.* 2001 (Comp Funct Genomics 2:80, mould-mitochondrial table 4).	Primary-source anchoring; both references added to the manuscript and supplement bibliographies (§Bibliography).
MIS-08	Main-text §2.3.6 tRNA analysis population paragraph rewritten	Named “5 variant codes (tables 4, 6, 10, 15, 31)” and said *S. cerevisiae* was “not included in the primary 24-pairing analysis”	Correctly names primary tables 3 (yeast mito CUN→Thr), 6, 10, 12, 15, 22 (chlorophycean mito equivalent of table 16), and 26; names *Blastocrithidia nonstop* + 2 *Mycoplasmoides* species as mechanistic boundary cases excluded from the 24-pairing analysis; states that *S. cerevisiae* mito Thr appears in every MIS	The prior text was inconsistent with the 24-row provenance table and with the MIS enumeration; the corrected paragraph is generated from the same 24-row input.
MIS-09	Main-text §4.3 signal-driver list corrected	Said signal driven by ciliates, *Euplotes*, “*Blepharisma* and *Mycoplasmoides*”	Signal driven by UAR→Gln ciliates, UGA→Cys *Euplotes*, and *Blepharisma stoltei* UGA→Trp; *Blastocrithidia* and *Mycoplasmoides* are excluded boundary cases (single-tRNA anticodon-stem or base-modification routes without gene-copy expansion)	*Mycoplasmoides* is outside the 24-pairing set precisely because its mechanism does not produce a copy-number signal; naming it as a driver reversed the evidentiary role.

ID	Change	Old value	New value	Justification
MIS-10	Supplement Table S11 gains a companion row-level provenance table	Only analysis-input columns visible	New Table S11b (Row-level provenance) with compartment, NCBI table id, source class, and assembly accession or primary citation for each of the 48 organism-slots; machine-readable CSV `output/tables/T-trna-24row-provenance.csv` extended with `variant_source_full` and `control_source_full` columns	Aggregate source-class counts alone were insufficient to audit each 2×2 table; the audit required row-level provenance.
MIS-11	Limitations, Conclusion, and README	"(suggestive)"; "p = 0.045"	"(exploratory)"; "median MIS p = 0.037, worst 0.104, topology-breaking subset p = 0.387"	Same reclassification as MIS-03, propagated to every venue that quotes the tRNA numbers.

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§NULL — Coloring-optimality null model

Context. The initial implementation labelled the primary coloring-optimality null "Freeland–Hurst block-preserving". The implemented ensemble is a *quartet-pattern shuffle* that holds each mobile first-two-base quartet AA-pattern atomic and permutes them across the 14 mobile quartet slots. The classical Haig–Hurst 1991 / Freeland–Hurst 1998 null described in the code-optimization literature is a different ensemble: it holds the 20 sense-codon families fixed and permutes the 20 amino-acid labels uniformly across the 20 families. The audit identified this as a naming discrepancy rather than a computational error — the quartet-pattern shuffle is a valid stringent null, but it should not be named after Haig and Hurst — and separately corrected the Monte Carlo tail convention.

Preservation properties. The quartet-pattern shuffle preserves each quartet's internal four-position amino-acid split shape (whether it is a (4), (2, 2), (3, 1), or singleton block) and each named amino acid's total codon count; it permutes which codons carry those labels. It does NOT fix the codon-family partition. The classical Haig–Hurst amino-acid permutation DOES fix the codon-family partition; it does NOT fix per-named-AA codon count. The state-space count for the quartet-pattern shuffle is $\approx 14! \approx 8.7 \times 10^{10}$ (14 non-stop quartets are mobile; the 2 stop-containing quartets are held fixed), not $\approx 16!$ as previously stated.

Monte Carlo tail convention. `monte_carlo_null` and its three sibling functions previously counted null draws with $f < F$_obs rather than the standard lower-tail permutation-test convention $f \leq F$_obs (i.e., they excluded ties at the observed value). The Grantham null contains exactly one draw equal to F_obs = 13 477; conservative p_cons moves from 61/10 001 = 0.006099 (strict) to 62/10 001 = 0.006199 (ties included). The other four metrics have no ties and are unchanged. All eight quartet-pattern-shuffle + Haig–Hurst p-values pass Bonferroni at $\alpha = 0.05/8 = 0.00625$.

ID	Change	Old value	New value	Justification
NULL-01	Renamed implemented null throughout main text and supplement	"Freeland–Hurst block-preserving null"	"quartet-pattern shuffle null" with an explicit naming/method paragraph in §2.3.1 and full comparison in §S2.1	Attribution accuracy: the implemented ensemble is not the classical AA-permutation null.
NULL-02	Added classical Haig–Hurst 1991 / Freeland–Hurst 1998 AA-permutation null	Not implemented	New generator `_generate_random_code_haig_hurst_aa_permutation` in `src/codon_topo/analysis/coloring_optimality.py`; wired via `null_type="haig_hurst_aa"` in `monte_carlo_null()`; emitted for all	Comparison sensitivity ensemble; lets readers see the effect of the null-ensemble choice on displayed p-values.

ID	Change	Old value	New value	Justification
			four metrics at n = 10 000, seed 135325, in `output/coloring_optimality_hh_aa_null.json`	
NULL-03	Supplement §S2.1 preservation-property table added	Not present	New comparison of what each null holds fixed vs. randomises, with per-metric p-value comparison and state-space counts (14! vs 20!)	Makes the null-ensemble choice explicit and readable.
NULL-04	Monte Carlo tail convention corrected (audit surface)	`k = number of null draws with f < F_obs` in four `monte_carlo_null()` functions	`k = number of null draws with f ≤ F_obs` (ties at the observed value included); `p_conservative = (k+1)/(n+1)`; result dict gains `n_at_or_below_observed`, `n_strictly_below_observed`, `n_ties_at_observed` fields	Standard lower-tail permutation-test convention. Only Grantham has a tie at F_obs; other metrics unchanged.
NULL-05	Main-text §2.3.1 methods rewritten for tail convention	"the number of null scores *below* the observed F"	"the number of null scores satisfying F_null ≤ F_obs; ties at the observed score are included in k"	Same convention as the corrected implementation.
NULL-06	Main-text §2.3.3 ρ-sweep literal updated	p_ρ=0 = 0.006, k = 60	p_ρ=0 = 0.0062, k = 61 (60 strict + 1 tie); other ρ values unchanged (no ties)	Same convention.
NULL-07	Main-text Fig 2A caption resolution increased	Caption rendered p at 3 significant digits (0.006)	Caption renders at 4 significant digits (0.0062), matching the figure's baked-in label	Prevents caption/figure mismatch given the Bonferroni threshold 0.00625 is close to the observed value.
NULL-08	Abstract and §3.1 Grantham p literals updated	Grantham p = 0.006	Grantham p = 0.0062 (four-digit display; important because the Bonferroni-corrected threshold at α = 0.05/8 = 0.00625 is close)	Editorial finding E13.
NULL-09	CLI option surface updated	`--null=freeland_hurst` (default) or `class_size`	Adds `--null=haig_hurst_aa`; help text clarifies "freeland_hurst = quartet-pattern shuffle; haig_hurst_aa = classical AA-permutation"	Makes both nulls reproducible from the command line.
NULL-10	§S18 cross-family multiplicity narrative rewritten	Placed raw p-values, within-family BH-FDR results, permutation p-values, and AICc gaps in one cross-family Bonferroni list	Recast as a descriptive conservative-threshold sensitivity: primary/smallest p-values reported against a common conservative reference threshold α^* = 0.05/8 = 6.25×10^{-3}; conditional-logit contribution reported as the M1⟶M3 likelihood-ratio test (χ^2 = 110.4 on 1 df, $p \ll 10^{-20}$) with the	AICc is not a p-value and cannot "survive" an α threshold; the reported quantities are non-exchangeable and do not form a single family-level list.

ID	Change	Old value	New value	Justification
			accompanying ΔAICc reported as a descriptive model-selection statistic; within-family corrections retained	

§H34 — H(3,4) clade-exclusion overclaim (main-text §3.4)

ID	Change	Old value	New value	Justification
H34-01	Main-text §3.4 clade-exclusion global threshold	"The depletion is highly significant in every regime ($p < 10^{-5}$)"	"Under the primary $H(3,4)/\Delta\beta_0 > 0$ definition all seven exclusions satisfy $p < 10^{-3}$ (largest $p \approx 2.03 \times 10^{-4}$, from excluding all metazoan mitochondria); under the sensitivity Q_6/new-disconnection definition all seven satisfy $p < 10^{-5}$"	Under the primary $H(3,4)$ adjacency, 2 of 7 regimes exceed 10^{-5} (ciliates 3.96×10^{-5}, metazoan mitochondria 2.03×10^{-4}); the $p < 10^{-5}$ statement is true only for the Q_6/new-disconnection sensitivity.
H34-02	Table S1 (claim hierarchy) topology-avoidance row aligned	"Clade-exclusion sensitivity (7 regimes): all $p < 10^{-5}$"	"Clade-exclusion sensitivity (7 regimes): all $p < 10^{-3}$ under $H(3,4)/\Delta\beta_0 > 0$ (largest $p \approx 2.03 \times 10^{-4}$); all $p < 10^{-5}$ under Q_6/new-disconnection"	Same fix in the registered claim hierarchy.
H34-03	Supplement §S9 (H(3,4) clade-exclusion table) retained	7-row table already present	Referenced from the corrected main-text §3.4 statement	Provides the underlying values that support the corrected statement.

§PARTIAL — Metric partial-correlation count

ID	Change	Old value	New value	Justification
PC-01	Main-text §3.1 count	"five of ten pairs become approximately uncorrelated after adjustment ($\|\rho\_partial\| < 0.1$)" naming Grantham~PR, PR~KD, PR~ProtSub	"four of ten pairs" naming Grantham~PR (+0.03), Miyata~ProtSub (−0.06), PR~KD (+0.01), PR~ProtSub (+0.08); the descriptor "a descriptive point-estimate statement, not a test of statistical independence" added	Table S19 lists exactly four pairs with $\|\rho\| < 0.1$; main-text and CHANGELOG count corrected.
PC-02	CHANGELOG partial-correlation line	"five of ten"	"four of ten"	Same.
PC-03	README §Coloring optimality	"five of ten pairs are statistically independent"	"four of ten pairs are near-zero after adjustment … (a descriptive point-estimate statement)"	Same.

§NOTATION — Topology-graph notation and quartet-null description

ID	Change	Old value	New value	Justification
N-01	Main-text §2.3.4 graph notation	“let G_a^1 denote the subgraph induced on codons assigned to *a* with edges between codons at binary Hamming distance ≤ 1” (Q_6-only)	“for each amino acid *a* and each adjacency $A \in \{Q_6, H(3,4)\}$, let G_A^a denote the subgraph induced on codons assigned to *a* with edges given by adjacency A; the topology-breaking indicator $\Delta\beta_0(G\_A\text{^}a) > 0$ is defined separately for each adjacency; the primary test uses A = H(3,4)”	Prior notation defined only the Q_6 variant while declaring H(3,4) primary; the corrected definition indexes the graph by adjacency.
N-02	Main-text §2.3.1 quartet-null description	“This null preserves both the synonymous codon contiguity (wobble degeneracy) and each quartet’s internal split structure”	“The null preserves each quartet’s internal four-position amino-acid split shape … and each named amino acid’s total codon count, while permuting *which* codons carry those labels; no quartet is broken up, but amino-acid labels move between quartet slots”	The prior phrasing suggested named-amino-acid to codon-set membership was fixed; the exact preservation properties are as stated.
N-03	Main-text §2.1 encoding-sweep sentence	“All encoding-dependent results are tested across all 24”	“The two encoding-dependent analyses in this work (cross-metric coloring optimality and the Q_6 topology-avoidance depletion) are evaluated under all 24 bijections; the M1–M4 conditional-logit models and the primary H(3,4) topology-avoidance test are encoding-invariant by construction and are not swept”	Prior sentence overstated the scope of the encoding sweep; only two encoding-dependent analyses are swept.
N-04	Abstract encoding-robustness wording	“robust to alternative topology definitions and base-to-bit encodings”	“The H(3,4) result is stable by construction; the Q_6 decomposition is representation-specific and fails to show depletion under 8 of 24 base-to-bit encodings, so we report H(3,4) as the primary test and Q_6 as a sensitivity”	Distinguishes the encoding-independent H(3,4) result from the encoding-dependent Q_6 decomposition.

§FRAMING — Discussion framing

ID	Change	Old value	New value	Justification
F-01	Main-text §4.2 per-table framing	“26 of 27 tables … suggesting that codon reassignment events are constrained to preserve error-minimization”	“26 of 27 tables remain low-cost relative to their own quartet-shuffle ensembles. Because the per-table null lacks distance-matched draws (§S8), this result is *compatible* with preservation of error-minimization during reassignment but does not identify independent variant-specific optimization”	The per-table null lacks distance-matched draws; the result is compatible with preservation but does not evidence independent optimization.
F-02	Supplement §S6 (IIA) framing	“the explanatory–rather–than–predictive framing makes IIA tolerable for our purposes”	“We use the model for explanatory comparison, not as an unconditional endorsement of IIA: IIA remains an untested structural assumption, the restricted-candidate refits are sensitivity analyses rather than evidence that IIA is harmless, and coefficients and likelihoods could shift under a mixed-logit relaxation”	Aligns §S19.1 with the more careful §S6 caveat that IIA is untested.
F-03	Supplement §S22 Slavov event-level framing	Event-weighted binomial $p < 10^{-100}$ presented as an “empirical signature”	Frequency contrast (65.0% vs 39.5%) framed as the primary descriptor; the event-weighted binomial p-value is labelled a descriptive naive summary that ignores clustering by substitution type, gene, sample, and measurement opportunity; the substitution-type analysis (OR 1.17, p = 0.26) is placed immediately beside it	Harmonises the supplement with the main text’s more careful phrasing.
F-04	Main-text Limitations 22-pair sentence	“Assembly-fragmentation robustness (removing two genomes with >10% pseudogene fraction yields Stouffer p = 0.041 on the remaining 22 pairings) is reported in Supplement §S10”	“Per-genome tRNAscan-SE pseudogene fractions vary across the 18 verified assemblies and are reported in Supplement §S10 (Table S12, *Pseudo* column); we do not report a formal fragmentation-based sensitivity test in this manuscript”	The prior sentence pointed to §S10 for a fragmentation-based sensitivity that the supplement did not document; the corrected sentence removes an unsupported claim.

§CLAIM — Claim-hierarchy definitions

ID	Change	Old value	New value	Justification
C-01	Main-text §2.6 SUPPORTED definition	"SUPPORTED (passes rigorous null, $p < 0.01$)"	"SUPPORTED (meets its prespecified claim-specific decision rule under the stated null or model and multiplicity control); EXPLORATORY (descriptive or hypothesis-generating; does not meet the relevant confirmatory gate); REJECTED; FALSIFIED; TAUTOLOGICAL — Each claim's decision rule is stated in the Justification column of the full hierarchy in the supplementary materials"	The prior blanket $p < 0.01$ rule did not fit the BH-FDR-based per-table claim; a per-claim decision rule is used.
C-02	`claim_hierarchy.py` SUPPORTED docstring	"Passes rigorous null; cite as finding"	"Meets its prespecified claim-specific decision rule under the stated null or model and multiplicity control"	Same.
C-03	Fig 5C caption + status labels	Labels "Verified / Tested / Pending / Falsified / Tautological / Exploratory" without cross-reference to Table S1	Labels renamed to registered categories where possible ("Supported / Rejected / Pending / Falsified / Tautological / Exploratory"); caption explains the per-prediction vs per-claim distinction	Aligns Figure 5C with the registered claim hierarchy.
C-04	Fig 5A axis label	"Feasibility score"	"Structural-preservation index S (visualization-only)"	Aligns Fig 5A with the Methods §2.5 rename to "Structural-preservation index (visualization-only)".

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§BIB — Bibliography additions and citation cleanups

ID	Change	Justification
B-01	Added `@bonitz1980` (Bonitz *et al.* 1980, PNAS 77:3167, doi:10.1073/pnas.77.6.3167) to `output/references.bib`; cited in §2.3.6 as the primary source for the *S. cerevisiae* mitochondrial tRNA vector.	Primary source for the corrected yeast-mito tRNA counts.
B-02	Added `@kerscher2001` (Kerscher *et al.* 2001, Comp Funct Genomics 2:80, doi:10.1002/cfg.72) to `output/references.bib`; cited in §2.3.6 as the primary source for the *Y. lipolytica* mitochondrial tRNA vector.	Primary source for the corrected *Y. lipolytica* mito tRNA counts.
B-03	Added explicit `@atchley2005` in-text citation at the Serine/Factor 3 exploratory-observation paragraph (main-text §3.7).	Prior text used $F\_3 = -4.760$ without citing the Atchley source.

ID	Change	Justification
B-04	Retained existing `@chan2016` GtRNAdb citation and referenced it at the GtRNAdb-sourced organism-slots in §2.3.6.	Replaces bare URL.

§AI — Generative-AI-assisted code disclosure

ID	Change	Justification
AI-01	New Methods subsection §2.6 "Software, validation, and reproducibility" added, describing Python 3.11 / R 4.4 stack, the use of generative-AI systems to scaffold portions of analysis and figure-generation code, human review, version control, unit / regression / property-based test coverage (>400 pytest tests), independent numerical anchor-value checks, and confirmation that no AI system generated or altered research data, inferential outputs, or figures.	Elsevier's current journal policy calls for Methods-level disclosure of AI-assisted code development in addition to the end-of-manuscript declaration.

§FIG — Figure fixes carried into this release

ID	Change	Justification
FIG-01	Figure 3B: added deterministic pre-jitter so that coincident (age, ε) markers render as visually distinct points in both PNG and PDF renders	Prior versions had visible jitter in the PDF render but not the PNG; the fix makes both consistent.
FIG-02	`save_figure()` in `theme_codon.R` now emits PNG + PDF + TIFF (LZW-compressed) for every figure.	Elsevier production requires TIFFs for production upload.

§DELIVERY — Delivery inventory (unified)

The single authoritative delivery inventory for this release, sorted by recipient use. Every entry is hashed in `MANIFEST.sha256`.

File	Purpose
`manuscript.pdf`	Main manuscript, Typst build; figures embedded at 300 DPI; PDF metadata (title, authors, keywords, date) written to the Info dictionary.
`supplement.pdf`	Supplement, Typst build; PDF metadata written to the Info dictionary.
`elsevier_response.pdf`	Response letter accompanying this delivery.
`corrections_ledger.pdf`	This document, rendered as a PDF.

File	Purpose
`submission_bundle.zip`	Editorial-Manager-ready bundle: `manuscript.tex` + `supplement.tex` (bibliography inlined via `\begin{thebibliography}` for self-contained pdflatex compilation) + `elsarticle.cls` + 300-DPI PNGs and TIFFs for every figure + `references.bib` + the compiled `manuscript.pdf` / `supplement.pdf`.
`submission_em_v0.6.1/`	Unpacked source of `submission_bundle.zip` for reviewers preferring an unarchived layout.
`CHANGELOG.md`	Release notes: consolidated v0.6.1 entry for the pre-publication QA/QC pass.
`2026-08-14-corrections-ledger.md`	Markdown source of this ledger.
`MANIFEST.sha256`	SHA-256 hashes of every file above. Verify with `shasum -a 256 -c MANIFEST.sha256` inside the folder.

Release tag on the public repository: `v0.6.1` (annotated).

§BUILD — Canonical build command

One canonical command reproduces every displayed value from a clean clone of the public repository at tag `v0.6.1`:

```
git clone https://github.com/biostochastics/codontopo && cd codontopo
git checkout v0.6.1
pip install -e ".[dev]"
bash scripts/build_publisher_release.sh
```

`scripts/build_publisher_release.sh` runs the analysis pipeline (`codon-topo all --seed=135325 --n=10000`), the provenance-emit and table-generation scripts, both R figure scripts, and the two Typst PDF compilations, in that order. The same four-line command block is referenced verbatim in the Elsevier response letter, `CHANGELOG.md`, main-text Data-and-code availability, supplement §S24, and `README.md`. Python dependency versions are pinned in `requirements.lock`; a reference `Dockerfile` provides a container floor.

§UNCHANGED — What was audited and NOT changed

- **The four SUPPORTED claims.** Cross-metric coloring optimality (all four code-independent metrics below α, *Grantham* $p^* = 0.0062$ after tie correction), per-table preservation (11 of 12 informative-distance variants below BH-FDR $\alpha = 0.05$), ρ-robustness ($p \le 0.006$ at every ρ), and $H(3,4)$ topology-avoidance depletion (hypergeometric $p = 1.28 \times 10^{-6}$; RR ≈ 0.32) all retain SUPPORTED status.
- **The FALSIFIED KRAS–Fano result** and the three REJECTED algebraic conjectures (Serine min-distance-4 invariant, PSL(2, 7) linear irrep, holomorphic embedding). Status unchanged; wording narrowed to reflect what the argument establishes.
- **The final claim-hierarchy total (15).** Category count: 4 SUPPORTED / 5 EXPLORATORY / 1 FALSIFIED / 3 REJECTED / 2 TAUTOLOGICAL, as in the accepted version.
- **All figure captions and legends** except those explicitly listed in §FIG.
- **All bibliography entries** except those in §BIB.

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